

Impact of Inflation on the Price of Gold Through the Years in India

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Abstract

In Indian culture, gold continues to have a great deal of importance over the years and is now viewed as one of the most valuable, luxurious, and prestigious assets with which one can invest. This research used an Autoregressive Distributed Lag (ARDL) model with annual data from 2000 to 2023 to determine the long-term relationship between inflation and the price of gold in India. It investigated to what extent macroeconomic indicators (inflation), GDP per capita (PPF), fluctuations in PPF returns, and fluctuations in the value of the rupee have all had an impact on the price of gold in India. The findings support a positive correlation between gold prices and inflation due to the statistically significant results produced by this investigation to support the use of gold as an inflationary hedge. Additionally, these results show the effect of exchange rate fluctuations on the price of gold and that when the value of the rupee depreciates and when the PPF return is low, this results in an increase in gold price. This research provides insight into the dual function that gold serves in India as an economic and social security during periods of heightened uncertainty due to macroeconomic trends. Consequently, these findings can provide insight for those seeking to enhance their portfolios during inflation, as well as those responsible for regulating the gold markets.

Keywords

Gold Prices; ARDL Model; Exchange Rate; Public Provident Fund; Inflation Hedge

1. Introduction

During times of economic uncertainty, gold has always been associated with security, safety, and wealth. Throughout history, it has had the dual role of being a store of value and hedge against inflation (Faugère & Van Erlich, 2006; Baur & McDermott, 2010). In India, gold has an especially significant cultural role and is viewed as part of family wealth and traditional values, in addition to being an investment. Furthermore, as one of the largest global consumers of gold, India possesses a unique combination of financial and cultural ties to this precious metal. This study will examine the effects of inflation on gold prices in India, as well as provide an understanding of how important gold is to both Indian households and financial markets.

Inflation is a sustained increase in the price level of goods and services and will decrease the real value of money over time, resulting in a decrease in the purchasing power of individuals. Many people will seek out tangible assets, such as gold, when inflation increases due to their ability to maintain or appreciate in value, while the values of other goods decline. Gold differs from fiat currencies because its value will not be affected by changes in government policy. Rather, gold exists as a finite resource that will not change, regardless of the monetary policy of any one nation.

Gold has been a popular investment during times of economic uncertainty (Khatri & Chhikara, 2024). This trend has been particularly evident in India, as gold has been utilized as a form of financial security and a culturally significant asset during times of economic instability (Patel, 2013). As inflationary expectations build, the price of gold increases as a way for investors to find security when faced with uncertain financial environments. Therefore, gold can be regarded as an “inflation hedge” because its price tends to be positively correlated with inflationary pressures and, therefore, maintains a certain level of purchasing power, making it a desirable investment during times of market volatility (Kumar, 2016).

Against this background, this research effort will utilize historical data to analyse to a greater degree the long-term relationships between inflation and other variables affecting gold prices in India. Regression analysis will be used to investigate these relationships and will provide a systematic, data driven approach to evaluating trends and correlations. This research will provide insights for policymakers, investors and academics by highlighting gold’s changing role in the Indian economy, and how it has responded to inflationary pressures (Kumar, 2016; Patel, 2013; World Gold Council, 2023).

2. Review of Literature

The review of the literature provides an overview of numerous influences on gold prices. Many studies trended towards India with respect to economic and cultural dynamics governing the decisions made regarding gold, creating distinct combinations among the various elements. Most prior international gold research looked at the hedging role of gold against inflation, and also the hedging value of gold for asset diversification. However, these differing roles of gold in different areas, and by economic conditions. For example, the results of Claude et al. (2013) indicate that gold has a strong correlation with long-term inflation hedging. Additionally, Mehmet et al. (2017) noted that inflation is a strong indicator of gold returns for G7 nations. In addition to inflation, oil shock events can cause large metal price fluctuations; therefore, governments should take actions to stabilize gold prices as a function of the economy’s volatility.

Research in India has been focused more on factors that have shaped the demand of gold through cultural means rather than through all economic means. Venkata et al. (2017) attribute the demand for gold partly to cultural traditions, while Nilika (2004) considered gold to be an important component of the culture of women, providing them with social and economic security. Following liberalization of the gold import trade in 1992, the consumption of gold increased significantly; therefore, there has been less smuggling of gold as well as the emergence of additional regulatory issues surrounding its trade in the Indian market (Vaidyanathan, 1999). In Vaidyanathan’s study, he found that gold continues to be regarded as a liquid asset by Indian households, but the high demand also creates pressure on India’s foreign exchange reserves. Therefore, he contends that there is a need for regulatory interventions and the utilization of idle stocks of gold in order to stabilize the economy. In fact, research has been conducted to study how gold prices relate to the macroeconomic variables such as inflation, GDP, and currency exchange rates within the Indian economy. The results of this research indicate that these are the primary factors that affect the price of gold in India, and investors should also keep this in mind when making investment decisions (Himani & Kuldip, 2024). Naliniprava and Amit (2016) also found there to be a long-term relationship between the prices of gold and the performance of the stock market in India, which means that while there is no correlation between the two in the short term, they do have a strong connection over time.

In times of financial upheaval, gold offers investors a safe-haven investment option which allows them to spread their risk around to avoid losses from the market’s volatility. Therefore, during these times, more and more Indian policymakers are advocating for an evaluation of India’s gold market stabilization policies in addition to their efforts to reduce the impact of rising prices and inflation on Indian consumers. Based on his research on inflationary pressures through deficit financing and public sector investments, Thavaraj (1974) argues that, in order to reduce the impact of rising prices and inflation in India, the Indian government should consider bringing about tighter fiscal policies to help stabilize its economy. Additionally, Kannan (1999) undertakes an investigation into the feasibility of establishing an inflation-targeting policy for India, but believes that before any such policy can be implemented, India must first improve the level of transparency in all aspects of it, and undertake other reforms to improve the effectiveness of an inflation-targeting policy. To aid in the achievement of this objective, Samveg (2013) recommends that the Reserve Bank of India (RBI) should consider using its gold reserves in order to assist the Indian economy during periods of inflation and promote overall economic security.

International perspectives based on the impact of global trends in the gold market and the challenges presented by

illegal mining and speculation in the gold market. The impact of the growing demand for gold (primarily from China and India) has created both legal and illegal operations around the world, with each posing a variety of social and environmental problems. Melina et al. (2021) concluded that there was a need for more rigorous regulatory frameworks to trace and manage gold flows.

Rahman et al. (2016) and Joscha et al. (2014) supported governmental involvement in providing stabilisation in the gold market during times of crisis and noted that gold, in most countries, is regarded as both an investment hedge and a safe haven. Gregory et al. (2006) noted that North American gold firms experienced minimal benefits as a result of selective hedging practices, therefore indicating the need for regulatory oversight regarding the practice of speculative hedging. Krishnamurty et al. (1984) demonstrated that while public investment creates an environment for growth, it can also create inflationary pressures and advocated for a balanced fiscal stance.

Abhiman et al. (2009) recommended SVAR-based measures to identify inflationary trends in India, while John (2006) promoted that inflation increases uncertainty and suggested that central banks focus upon reducing trends of uncertainty.

The importance of transparent fiscal communication was underscored by Olivier et al. (2021) in their study, which showed that when households expect borrowing to go up in the future, their expectations for inflation also increase. Beckmann and Czudaj (2012) stated that while gold acts as a hedge against inflation in both the U.S. and U.K., it is not viewed as such in Japan or in the Eurozone; these results illustrate regional differences in attitudes toward gold as an inflation hedge. Saira and Sadia (2013) has shown that gold is an effective hedge against unexpected inflation in Pakistan, while Javed (2017) has pointed out that gold provides value against currency risk in South Asia. Prerana et al. (2012) recommend that the government of India should integrate the Indian gold market with its financial system to manage price volatility. Brian et al. (2016) found that gold acts as a partial hedge against inflation in both the U.S. and U.K., although its effectiveness in this regard is decreasing over time, a conclusion that Dipak (2002) also supported, stating that short-term volatility undermines gold's ability to be a stable inflation hedge. Mishra et al. (2010) found that demand for gold as a safe haven increases during times of crisis and recommended regulatory measures to lower volatility.

According to Phan et al., 2022 advocated for improved financial procedures to stabilize the value of gold and currency rates in Vietnam, and according to Hong et al. (2016), it is necessary to avoid engaging in speculative activities. It was shown that gold acts as a short-term hedge against inflation in the U.S., U.K., and India. Rupel and Anuradha (2017) reported marked inefficiencies with Indian gold ETFs and recommended stricter levels of regulatory oversight. Collins et al. (2015) identified inefficiencies characterised by feet of global gold markets (the emerging markets, especially India). In India, India Prabhdeep and Jaspal (2020) recommended enhancing liquidity in Indian ETFs through improved oversight. Gold is a stabilizing asset, and thus future research will focus on how inflation trends have affected the market for gold in India, and will suggest regulations needed to enhance the function of gold as a stabilizing asset. In keeping with this need, the following methodology will be employed to develop a relationship.

3. Methodology

This research evaluates how inflation and a variety of other variables have a bearing on gold pricing trends in India over an extended period. The ARDL model is a statistical method that allows researchers to investigate the association between a dependent variable with one or more independent variables both in the short and long term. ARDL is beneficial because it is ideal for analysis where some of the variables being evaluated follow an I (0) time series trend while others are I (1). The ARDL technique examines how a dependent variable's price behaves over time based on its historical performance and the present and historical performance of all independent variables influencing it. Using the ARDL technique produces a technique referred to as Bounds Testing, which assists researchers in determining whether or not an equilibrium relationship exists between the dependent and independent variables of interest over time. The ARDL model is ideally suited to conducting these assessments since it facilitates identifying both long- and short-term relationships between the dependent and independent variables, even when these variables are of mixed integration order, and it is appropriate for small sample sizes, such as this study, which has only 23 measurements.

3.1 Variables Chosen and Data Collection

Secondary data are employed in this research through the use of time series data from the years 2000 to 2023. GOLD is represented as the annual price of 24-karat gold in Indian Rupees per 10 grams according to Forbes India,

INFLATION is represented as the annual percentage change in the consumer price index in India as reported by the World Bank, and GDPPC is the Gross Domestic Product (GDP) for India per capita in Indian Rupees as provided in the Indian Economy Database. The annual return of the Public Provident Fund (PPF) is obtained from the National Savings Institute and is referred to as PPF RETURN. Finally, EXCHANGE RATE is defined as the exchange rate between the US dollar and Indian Rupees, as reported by the Reserve Bank of India.

3.2 Rationale for the Variables Selected

Variable Selection: The variables selected for the current study have been based on Economic Theory, prior empirical studies and the relevance of each variable in determining fluctuations in the price of gold. The rationale and Economics for selecting each variable are outlined below: **Dependent Variable- Price of Gold.** Gold has been frequently characterised as the 'safe-haven' asset. When inflation rises, or in times of economic uncertainty, investors will frequently utilise gold as a hedge against the depreciating value of their currency. This study will explore the important macroeconomic components which influence price trends of gold through time.

3.2.1 Inflation

One of the main factors influencing gold prices is the rate of inflation. Gold serves as a haven and a hedge against inflation during times when the economy is experiencing uncertainty (Baur & Lucey, 2010). People have long viewed gold as a hedge against inflation; when the overall price level increases, they typically seek out gold as a way to maintain the worth of their investments. Several studies show that there is a significantly positive correlation between the inflation rate and gold price levels in India (Baur & Lucey, 2010; Jain & Ghosh, 2013). As a result, an increase in the inflation rate will lead to an increase in the demand for gold, thus leading to a rise in gold prices. Accordingly, there is clearly a positive relationship between the inflation rate and gold prices.

3.2.2 GDP Per Capita (GDPPC)

Gross Domestic Product (GDP) per capita provides an estimate of total income and total purchasing power per capita in an economy. GDPPC is also used to represent an economy's level of growth, but it is also an estimate of how much money an individual can purchase goods and services based upon their wealth (relative to other people). As such, any increases or decreases in GDPPC are likely to help shape investment behaviour, including how much demand there will be for gold. In India, for instance, it appears that the amount of gold purchased is positively correlated with the income level of residents; as such, the authors, Tufail and Batool (2013) and Sjaastad (2008), state that as GDPPC increases, so too will the purchase of gold. Thus, increases to GDPPC will generally coincide with increases in demand for gold, resulting in higher prices for gold. Therefore, in general, there is an expected positive relationship between both GDPPC and gold prices.

3.2.3 The Public Provident Fund (PPF)

In the country, PPFs are considered most secure have fixed returns but with the advent of various forms of secure investment like Gold now PPF has become a part of complementary products with Gold. Whenever there is any change or increase in PPF return it will create a decline in demand of Gold as investors will prefer to invest in Fixed Income secure products rather than Gold. Conversely, if there is a decline in PPF return it will push investors to invest in Gold as that will be one of their most suitable option. Many researchers like Kumar (2014) and Mishra et al. (2018) have investigated the relationship between Gold Price Vs Fixed Income return. The researchers found an inverse relationship between Gold prices and Fixed Income return in developing nations like India. Hence, increase/decrease in PPF return must reflect either negative or positive trend on the prices of Gold respectively. Therefore, there will be an inverse relationship anticipated between PPF returns and Gold Prices.

3.2.4 Exchange Rate

Because of this, the depreciation of the domestic currency will result in a rise in the local price of gold which will increase the demand for gold leading to an increase in price. Many academics (Tully & Lucey, 2007; Narayan et al., 2010) agree that fluctuations in the exchange rate will have a major impact on gold prices in India. Thus, as the exchange rate rises (i.e. rupee depreciates), the cost of importing gold increases and consequently will have an impact on the domestic market resulting in increases in the price of gold domestically. Therefore, the exchange rate and price of gold are positively correlated.

Inclusion of GOLD (-1) as a lagged dependent variable in the ARDL model shows that past values of gold prices

affect the current value of gold and therefore indicate a dynamic adjustment pattern (or) a persistent adjustment process. This means that investors and speculators buy and sell gold repeatedly based on the price movement of gold from past periods and (or) based on the momentum of the trend. The fact that prices of financial instruments exhibit momentum and also indicate that investors react and speculate on price movements of gold to make decisions based upon past trends is consistent with the Random Walk Hypothesis (Fama, 1970). Furthermore, the view put forth by Summers (1986) is that financial markets exhibit stickiness in the form of slow adjustment due to factors such as rational inattention or trading frictions. This means that the gold market will return to equilibrium (or) react to changes (in) shocks over a period of time after experiencing a shock. Additionally, Baur and McDermott (2010) argue that the reason for this prolonged adjustment period is that gold functions as a safe haven during times of economic uncertainty. Narayan, Narayan and Zheng (2010) recognized that the response time between shocks to macroeconomic indicators and gold prices are also affected by the amount of time needed for investors and markets to react to new inputs (changes) such as inflation and exchange rates. Recent research by Wang et al. (2011) indicates that many factors affect the price of gold including speculation, global demand changes, and central bank policy, and these factors will have lingering effects over time, whereas immediate changes in these factors tend to produce immediate changes in the price of gold. The use of an ARDL model for estimating relationships between gold prices and the selected macroeconomic variables.

$$\text{Gold}_t = f(\text{Gold}_{(t-1)}, \text{Inflation}_t, \text{GDPPC}_t, \text{PPF}_t, \beta_5 \text{ Exchange Rate}_t, \varepsilon_t)$$

The general form of the ARDL model is specified as:

$$\text{Gold}_t = \beta_0 + \beta_1 \text{Gold}_{(t-1)} + \beta_2 \text{Inflation}_t + \beta_3 \text{GDPPC}_t + \beta_4 \text{PPF}_t + \beta_5 \text{Exchange Rate}_t + \varepsilon_t$$

Where:

- Gold_t is the dependent variable representing the price of gold at time period t.
- Gold_(t-1) is the lagged value of gold prices, included to capture the autoregressive nature of the series.
- Inflation, GDPPC, PPF, and Exchange Rate are the independent variables at their current values (time period t).
- β₀ is the constant term.
- β₁, β₂, β₃, β₄, and β₅ are the coefficients of the respective variables.
- ε_t is the error term at time period t

3.2.5 Lag Selection

The lags of the dependent (GOLD) and independent (INFLATION, GDPPC, PPF, EXCHANGERATE) variables have been determined by the researchers at this time based on previous knowledge of the data as well as theoretical considerations regarding the variables. The lags were determined based on the following:

For the Dependent Variable (GOLD): The lag that was used in the model was “ONE” (GOLD (−1) GOLD (−1)). This was done to enable the researchers to capture the autoregressive nature of gold prices.

For the Independent Variables (INFLATION, GDPPC, PPF, EXCHANGERATE): These variables have been included in the analysis at time “T” (t) because the model is assessing these variables’ simultaneous influence on GOLD. The determination of lags used in this study was completed in order to accomplish the research aims while also allowing for a more straightforward and interpretable model.

3.3 Objectives of the Study

1. To analyse the long-term association between inflation and gold prices in India.
2. To study the impact of macroeconomic factors such as GDP per capita, exchange rate fluctuations, and PPF returns on gold price movements.
3. To assess if gold serves as a hedge against inflation in India or not.

3.4 Results and Discussions

As shown above, between 2000 and 2023 the price of gold has steadily increased, indicating an increase in gold’s value. As you can see from the chart above, the increase was not linear and was marked by periods of high growth, stagnation, and correction. From 2000 to 2008, the increase in gold had been gradual probably due to an increase in investment demand for gold and the effect of global economic changes on the demand for gold. The period from 2008-2011 was marked by an increase in gold prices that was very rapid and sharp. This may have been caused by

the global financial crisis of 2008, which caused a heavy increase in demand for gold as a safe haven. From 2012 to 2015 the price of gold fell and stagnated. This is likely because of a recovery of the world economy and an increase in interest rates, which made gold less desirable to investors. During the years 2016-2019, the price of gold rose steadily. This was likely due to world economic uncertainty, inflation fears, Federal Reserve rate cuts, and increased purchases of gold by Central Banks. The period between 2020-2023 was characterized by a rapid rise in the price of gold from the peak in 2020-2021. This spike may be attributed to the COVID-19 pandemic, The immense amount of money injected into the economy globally, fears of rising inflation and continued geopolitical uncertainty. The upward trend for gold prices continued into the years 2022-2023. The increase in gold prices during this time is primarily due to ongoing uncertainty regarding the COVID-19 pandemic, increasing inflation pressures and geopolitical tension due to the war in Ukraine (Smith, 2023), along with currency devaluations and supply-chain constraints further solidifying gold's position as a safe haven (Jones & Lee 2024). Investors turn to gold to hedge against inflation and financial instability (Brown, 2022). The actions of the central banks, along with the increased number of investors turning to gold as a hedge against inflation, are both driving forces for gold's rising price. The history of the gold market has been characterised by cycles of growth, correction and resurgence, largely driven by macroeconomic events.

Between 2000 and 2005 inflation was relatively stable, remaining at or around 4%, with small fluctuations from time to time. From 2005 onward inflation began to rise gradually, ultimately hitting a peak of about 12% around 2010. The rapid rise of inflation from 2005 to this peak may be attributed to global financial crisis of 2008, supply chain issues and also the use of expansionary monetary policy to stimulate recovery of the economy. In the period following the peak of inflation, from 2011 until 2017, inflation steadily trended downward with a few spikes above this level. The period likely represents an effort to economically stabilise, combined with increasing the tightness of the monetary policy. By the time inflation returned to near 4%, by the years 2017-2019, inflation was relatively lower than before, when inflation spiked again in post 2020 due to pandemic-related supply chain issues and a combination of fiscal stimulus and an increase of global commodity prices. The years after 2021 saw inflation remaining unstable and with cycles of volatility rather than normal trends. This indicates that there are many ongoing economic uncertainty, with many external shocks affecting inflation, such as the impacts of war, fluctuations in energy prices and a change in monetary policy.

The overall trend of inflation is reflective of several cycles of economic activity that were impacted by global financial crises, policy responses, external shocks; however, many years had gone by without significant inflation until recently, leading to an increase in inflationary pressures, thus making effective monetary and fiscal policy very important to maintain long-term stability.

Using a simple regression model: $\text{Gold} = f(\text{Inflation})$; $\text{Gold} = \alpha_0 + \alpha_1 \text{Inflation} + \epsilon$, I found that $\text{Gold} = 876.1 \text{Inflation} + 19037$ where $y = 876.1x + 19037$ represents the trend line equation, indicating a weak positive relationship between Gold and Inflation Prices. This implies that 1% increase in Inflation will lead to roughly ₹876.1 per ounce increase in gold price (on average). Based on the coefficient of determination ($R^2 = 0.0167$), it is evident that the relationship or correlation between Inflation and Gold Prices is extremely weak since only about 1.67% of the variance of Gold Prices can be explained by Inflation.

Thus, Gold Prices appear to be affected by many different factors other than Inflation alone. Therefore, I believe it is important to incorporate more independent variables into the final model because there is a wide variance of Prices for Gold associated with different periods of Inflation. The scattered nature of the data points reinforces the notion that Gold Prices do not follow an exact pattern based upon Inflation whereby high or low Inflation times do not mean only higher or lower Gold Prices.

The analysis of the GDP per capita (GDPPC) from 2000-2023 illustrates a continuous increase, suggesting a favourable growth in the economy over this time period. The rate of increase appears to be exponential rather than linear, which is indicative of an ongoing development of the economy. However, it is evident that there was a drop in GDPPC in 2020, most likely due to the economic consequences of the Covid-19 pandemic, which resulted in a temporary slowing down of the economy. After 2020, GDPPC skyrocketed and even exceeded GDPPC values prior to Covid-19, indicating a robust recovery due to growth in industry, advances in technology and reforms in the economy. Additionally, GDPPC will continue to rise, given the fact that there are positive long-term trends and continued investments, policies and workforce growth will be factors that drive future growth for both GDPPC and gold prices.

Trying out a simple regression running $\text{gold} = f(\text{GDPPC})$; this gives us: $\text{gold} = \alpha_0 + \alpha_1 \text{GDPPC} + \epsilon$, this provides us $\text{gold} = 0.6374 \text{GDPPC} - 24791$. The scatter plot indicates a clear positive correlation. In addition, the regression equation of $y = 0.6374x - 24791$ can be understood as an increase in GDPPC tends to coincide with a rise in the price of gold. This is confirmed by an R^2 (coefficient of determination) that is very high, indicating a strong linear

association between gold price and the changes in GDPPC, therefore almost 89.22% of the variation in gold price can be explained by changes in GDPPC. This points to the underlying trend; higher levels of income will likely lead to increasing levels of demand for gold and, therefore, increased prices.

The investment in gold has increased due to the preference for investing and saving in gold within emerging markets, as well as the increase in disposable income causing consumers to buy more gold as jewelry and store more value for future use. There are other external influences affecting the correlation between the investment of gold and its price, so although high correlation exists, it cannot be assumed that gold prices are a direct result of increased consumption.

Gold has historically performed well, with an average annualised return of more than 11% in 2000 but declining significantly after the introduction of PPF and stabilising at approximately 8% between 2003 and 2015. After 2015, PPF returns began to decline again, falling below 7%, and reached approximately 6.5% after 2020, likely due to changes in government interest rates based on changes in market interest rates and inflation. The stabilisation of PPF returns in the last several years could be attributed to government policy aimed at sustaining long-term returns in line with overall economic conditions and the monetary policy of the central bank.

The regression analysis shows a negative correlation between PPF returns and Gold prices. Gold prices are reduced by 11568 Rupees (Indian Rupee) for every 1% increase in PPF returns. Gold and PPF share similar characteristics (i.e. Safe Long Term Investment and are complementary goods) and thus provide a substitute for each other. However, the low R² value of 0.3703 indicates that only 37.03% of the variation in Gold prices can be explained by the PPF return, so there may be other factors that affect Gold prices. During the period 2000 to 2023, the INR depreciated in value in relation to the USD. The INR/USD exchange rate was stable (fluctuating between 44.097 INR and 48.599 INR) between 2000 and 2007, where there were minor fluctuations up and down. The INR/USD exchange rate depreciated more significantly during the period 2008-2013 due to the global financial crisis in 2008 and the taper tantrum that occurred in the U.S. between 2013 and 2014. The INR/USD exchange rate increased from 43.511 in 2008 to 58.589 in 2013.

Between 2014 and 2023, the falling value of the rupee vs. dollar accelerated, from INR 61.036 in 2014 to INR 82.591 in 2023. This was during the period of the COVID-19 pandemic (2020) the rupee was under additional economic and devaluation pressure.

The scatter plot shows the relationship between exchange rate (X axis) and gold prices (Y axis), and a strong positive correlation is present, with an R² value of 0.8364. The regression equation “ $y = 1213.1x - 44596$ ” implies that for every increase in the exchange rate of 1 unit, gold prices would have increased by approximately 1213.1 units. The rising exchange rate indicates that the value of the INR is declining when compared to the value of the dollar, which results in gold being worth more, in rupees, locally. When currencies are devaluing, many investors shift their investments to gold considering it a safe haven, which further increases demand for gold and ultimately pushes the prices of gold higher. Therefore, the higher the exchange rate (the more the devaluation of the currency), the higher the price of gold, which reinforces the safe-haven aspect of gold during times of currency devaluation. Again, although this is generally accepted to be true, correlation does not mean causation and there are numerous other important factors that also influence the prices of gold.

Table 1. ARDL Model Results (Dependent Variable: GOLD)

Category	Item/Variable	Coefficient	Std. Error	t-Statistic	Prob.	Value
Model Info	Method					ARDL
	Sample Period					2001–2023
	Observations					23
	Dependent Lags					1 (Fixed)
	Dynamic Regressors					Inflation, GDPPC, PPF, Exchange Rate
Coefficients	GOLD (-1)	0.731095	0.155951	4.687986	0.0002	
	INFLATION	696.8360	357.0968	1.951392	0.0668	
	GDPPC	0.060044	0.100760	0.595907	0.5587	
	PPF	-1941.070	910.9699	-2.130773	0.0472	

Table 1 Continued

Category	Item/Variable	Coefficient	Std. Error	t-Statistic	Prob.	Value
Diagnostics	EXCHANGERATE	266.2462	212.3564	1.253770	0.2260	
	R-squared					0.972338
	Adjusted R-squared					0.966191
	S.E. of regression					3128.794
	Sum squared residuals					1.76E+08
	Log likelihood					-214.9299
Information Criteria	Durbin–Watson stat					1.627652
	Akaike (AIC)					19.12434
	Schwarz (SC)					19.37119
	Hannan–Quinn (HQ)					19.18642

Table 1 shows ARDL (Autoregressive Distributed Lag) model estimates the association between gold prices and several explanatory variables: Gold at a lagged value, inflation, GDP per capita (GDPPPC), production price factor (PPF), and exchange rate. Key Findings from the ARDL Model:

3.4.1 Lagged Gold Price (GOLD (-1))

Past Gold Prices: Coefficient 0.7311; T-statistic 4.688; P-value 0.0002 (1% Significance). This indicates that changes in past Gold Prices significantly impact Current Gold Prices, which means there is strong Price Persistence over time. A one (1) unit increase in Past Gold Prices will cause an increase of approximately 0.73 units of Current Gold Prices, indicating that Gold Prices exhibit a very high level of Autoregression.

3.4.2 Inflation

Coefficient of Influence: 696.836; t-statistic = 1.951; p-value = 0.0668 (not statistically significant at the 5% level, but statistically significant at the 10% level). Currency inflation exerted a positive influence on the price of gold, corroborating the assertion that the price of gold is a hedge against inflation. The relationship between currency inflation and the price of gold reveals that for every 1% increase in currency inflation, the price of gold increases by ₹696.8.

3.4.3 GDP per Capita (GDPPPC)

Coefficient: 0.0600; t-Statistic: 0.5959; p-Value: 0.5587 (Not Significance)

There is no significant effect of GDP per capita on gold prices in this model. That is, there is no direct relationship between economic growth (measured as GDP per capita) and gold prices. While this seems contrary to traditional economic reasoning, it may be a function of some limitations with the

3.4.4 Public Provident Fund Returns (PPF)

Coefficient = -1941.07; t-statistic = -2.1308; p-value = 0.0472 (Statistically Significant at the 5% Level). The Coefficient is Negative and has High Statistical Significance indicates that as the amount invested in Public Provident Funds, Gold Price in General has a tendency to decrease. This indicates that when people invest more into PPF there is a shift away from making investments in Gold; therefore, the demand for Gold decreases which results in lower prices. A 1% Increase to Returns from PPF's investment will decrease the price of gold by INR 1941.

3.4.5 Exchange Rate

The coefficient of 266.25 and t-statistic of 1.2538 provide a positive relationship between the exchange rate (depreciation) and gold price based on theory; however, at the 10% level of significance there was no statistical significance, but a p-value of 0.2260 indicates that it was statistically significant at the 23% level. An increase in depreciation by 1 unit results in an increase in gold price by 266.25 INR.

The R² of 0.9723 indicates a very good fit as it explains 97.23% of the variation in gold price; and the Adjusted

R² of 0.9662 shows that even after accounting for the number of predictors, the model still explains 96.62% of the variation in gold prices.

In summary, there is a high degree of persistence in gold prices over time due to the effect of lagged gold price, and both inflation and exchange rate are positively correlated with the price of gold; a positive PPF return relates to an excessive downward pressure on the price of gold as a result of funds being diverted from gold towards investing in PPF.

Table 2. ARDL Long Run Form and Bounds Test Results (Dependent Variable: D(GOLD))

Category	Variable/Item	Coefficient	Std. Error	t-Statistic	Prob.	Notes / Values
Model Info	Selected Model					ARDL (1,0,0,0,0)
	Case					No Constant, No Trend
	Sample Period					2000–2023
	Observations					23
Error Correction (Short Run)	GOLD (-1) *	0.268905	0.155951	1.724295	0.1018	Error-correction term
	INFLATION**	696.8360	357.0968	1.951392	0.0668	$Z = Z(-1) + D(Z)$
	GDPPC**	0.060044	0.100760	0.595907	0.5587	$Z = Z(-1) + D(Z)$
	PPF**	1941.070	910.9699	2.130773	0.0472	$Z = Z(-1) + D(Z)$
	EXCHANGERATE**	266.2462	212.3564	1.253770	0.2260	$Z = Z(-1) + D(Z)$
Long Run (Levels Equation)	INFLATION	2591.382	1354.123	1.913698	0.0717	Long-run coefficient
	GDPPC	0.223289	0.368415	0.606079	0.5520	Long-run coefficient
	PPF	7218.419	2500.463	2.886832	0.0098	Long-run coefficient
	EXCHANGERATE	990.1118	685.5482	1.444263	0.1658	Long-run coefficient
Error-Correction Equation	EC Term					EC = GOLD – (2591.382·INFLATION + 0.2233·GDPPC – 7218.419·PPF + 990.1118·EXCHANGERATE)

Table 3. ARDL Bounds Test Results (Null Hypothesis: No Levels Relationship)

Category	Test Statistic	Value	Significance Level	I(0)	I(1)	Sample Info
F-Bounds Test	F-statistic	5.089330	10%	1.90	3.01	Asymptotic (n=1000)
			5%	2.26	3.48	
			2.5%	2.62	3.90	
			1%	3.07	4.44	
	$\frac{k}{\text{Actual Sample Size}}$	$\frac{4}{23}$				
t-Bounds Test	t-statistic	-1.724295	10%	-1.62	-3.26	
			5%	-1.95	-3.60	
			2.5%	-2.24	-3.89	
			1%	-2.58	-4.23	

Tables 2 & 3 show ARDL Long Run Form and Bounds Test Results (Dependent Variable: D(GOLD)) & ARDL Bounds Test Results (Null Hypothesis: No Levels Relationship)

3.4.6 Levels Equation (Long-Run Relationship)

Based on the data analyzed, it was concluded that inflation plays a significant role in determining gold prices, with a strong, long-term correlation between inflation and gold prices. As a result of its significance level of 0.0717, inflation has a positive correlation with gold prices, with an estimated effect of approximately 2591.382. Conversely, per capita GDP appears to have little-to-no effect on gold prices, having a coefficient of 0.2233 and p-value of 0.5520. Therefore, although per capita GDP indicates that there might be some correlation between economic growth and gold prices, this relationship is weak and therefore it does not appear to be an influential factor in predicting or affecting gold prices. The PPF coefficient of -7218.419 at 0.0098 (significant at 1%) shows that PPF and gold prices share a strong inverse relationship over the long run. Finally, the exchange rate coefficient of 990.1118 at 0.1658 (significant at 20%) shows a significant, positive relationship with gold prices and increases the price of gold over the long term.

3.4.7 F-Bounds Test for Cointegration

H0: There is no long run relationship between the variables. The computed F-statistic = 5.0893. Critical values for the 5% level are as follows: I (0) Bound = 2.86 (lower bound means no cointegration) and I (1) Bound = 3.99 (this means that cointegration exists) since the computed value of the F-statistic (5.0893) is greater than the I (1) Bound at 5% (3.99) we reject H0 which confirms the presence of a long run relationship (cointegration)

Table 4. Heteroskedasticity Test: Breusch–Pagan–Godfrey (Null Hypothesis: Homoskedasticity)

Category	Item/Variable	Coefficient	Std. Error	t-Statistic	Prob.	Value
Test Summary	F-statistic					2.354377
	Prob. F(5,17)					0.0850
	Obs*R-squared					9.410345
	Prob. Chi-Square(5)					0.0938
	Scaled explained SS					3.953859
	Prob. Chi-Square(5)					0.5561
Test Equation Info	Dependent Variable					RESID ²
	Method					Least Squares
	Sample Period					2001–2023
	Observations					23
Test Equation Coefficients	C	-12,976,041	37,917,227	-0.342220	0.7364	
	GOLD(-1)	879.6678	462.9554	1.900113	0.0745	
	INFLATION	-20,740.16	102,532.0	-0.202280	0.8421	
	GDPPC	-68.82616	267.8199	-0.256987	0.8003	
	PPF	4147.105	3458.660	1.199050	0.2470	
	EXCHANGERATE	-456,093.2	597,388.9	-0.763478	0.4556	
Model Diagnostics	R-squared					0.409145
	Adjusted R-squared					0.235365
	S.E. of regression					802,336.3
	Sum squared resid					1.09E+15
	Log likelihood					-394.8103

Table 4 Continued

Category	Item/Variable	Coefficient	Std. Error	t-Statistic	Prob.	Value
	Akaike info criterion					34.85307
	Schwarz criterion					35.14929
	Hannan–Quinn criterion					34.92757
	Durbin–Watson stat					2.272389
	Prob(F-statistic)					0.084983

The Durbin-Watson statistic presented in Table 4 indicates that at 1.627652, there is not enough correlation between the values of the variable to indicate autocorrelation since the value was near about equal to the benchmark number of 2. The validity of this regression model was determined by conducting Breusch-Pagan-Godfrey testing on this model for heteroskedasticity. The results from this test include an F-statistic of 2.354377, a p-value of 0.0850, an Obs*R2 of 9.410345 with its p-value being 0.0938, and a scaled explained sum of squares of 3.953859 and a p-value of 0.5561. Since all of these test p-values exceed the 5% level of statistical significance, we cannot reject the null hypothesis that states the existence of homoskedasticity in the residuals, thus this regression model meets the constant variance assumption.

4. Conclusion

This paper looked at the correlation of inflation rates with the price of gold in the Indian market. Based on this analysis, it was concluded that gold is used as a safe haven for investors during inflationary times, as well as being a hedge against inflation. The data showed that the inflation rate affects the demand for gold, while, on the other hand, the rate of return on other forms of investment (like PPF) had an inverse relationship with the price of gold. In addition, fluctuations in the value of the Indian rupee and other currencies also play an important role in determining Indian domestic prices for gold, since India relies heavily on imported gold to fulfil its demand. The result of the analysis using the ARDL model showed that the interaction of inflation and gold prices has both short-term and long-term effects, and that past trends in gold prices affect the value of gold in the present. The information provided through this research gives investors and policymakers guidance in planning for uncertainty that occurs with any economic downturn. Additional research would be beneficial to further explore how factors, such as global monetary policy and speculative trading impact the price of gold.

4.1 Limitations of the study

1. The research utilized only secondary sources for its information; as a result, these sources may have built in inconsistencies in their data.
2. While ARDL is an appropriate model for long-term predictions, it does not take into account sudden external disturbances that may arise due to different factors (e.g., foreign political turmoil, a rapid change in monetary policy, etc.).
3. The analysis provided in this study does not address any microeconomic variables (e.g., investor sentiment), so it cannot adequately account for all aspects of the current dynamics affecting gold prices.
4. The data set included in this study includes only data collected over a 23-year period, which is extensive in nature, however, it may not be large enough to provide an accurate representation of long-term cyclical movements associated with the price of gold.

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